

What Is Remembered, What Is Recorded: Reasoning Continuity in Interrupted LLM-Assisted Work

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Resuming interrupted LLM-assisted decision work requires recovering not only prior content but where one’s reasoning stood. We operationalize this as Reasoning Position, a participant-endorsed checkpoint comprising judgments, rationales, constraints, uncertainties, and next actions. In a study with 33 participants and 252 reference propositions, unaided recall and frozen records overlapped only partially, 83 of 146 explicitly recorded propositions were not fully retained, whereas 19 of 58 insufficiently supported propositions were fully retained. These findings informed the Reasoning Memory Workspace (RMW), combining a recovery brief, reasoning cards, and a relational network. Unaided retention after interruption was 0.655, immediate post-support retention reached 0.905 with RMW under the same prompts, with smaller gains from prose summaries and self-authored notes. Our RMW illustrates a broader design direction for human–AI collaboration, moving beyond preserving conversations to helping users recover, supplement and advance prior reasoning.

CCS Concepts: • **Human-centered computing** → **Laboratory experiments**; *Human computer interaction (HCI)*; *HCI design and evaluation methods*.

Additional Key Words and Phrases: Task interruption and resumption, reasoning state recovery, LLM-assisted knowledge work, human-AI collaboration, interaction trace analysis, cognitive state externalization

1 Introduction

Complex analytical work rarely finishes in one uninterrupted sitting. People develop interpretations, weigh alternatives and refine judgments across meetings, messages and task switches. When they return, the documents are still there but resumption remains costly, because continuing a task takes more than retrieving its content [9, 29]. A returning worker must reactivate a suspended goal and rebuild the problem state that made the work actionable [1, 4, 16]. They may remember the conclusion they reached without recalling why they reached it, which constraints still bind, which alternatives they had already ruled out, what remains uncertain or what they meant to verify next.

LLM-assisted work makes this recovery need especially relevant. Conversations, documents, notes, and interaction histories persist across sessions, and model-side research continues to improve long-context access and multi-turn reliability [22, 27, 28, 40]. A returning user must still reconnect that information with the analysis they were developing [13]. This makes human reasoning continuity an interaction design concern, supporting should help users locate their prior position, inspect why it was held, and identify what remains to be done.

We call this intermediate state the **Reasoning Position**: a task-relevant position that a user can inspect and endorse at a checkpoint. Drawing on suspended goals, problem-state reconstruction, sensemaking, and metacognition [1, 4, 16, 35, 44, 51], we operationalize it through six components: **Goal**, **Current Judgment and Rationale**, **Active Constraint**, **Rejected Path and Rationale**, **Unresolved Uncertainty** and **Next Action**. Together, these components make the intended object of recovery explicit, including the qualifications and intentions that make a judgment actionable.

Persistent records support this recovery, though their coverage remains incomplete [36]: viewing a source does not establish belief, nor does an earlier discussion imply an active choice [5, 30]. We thus examine both post-interruption recall and the support provided by earlier traces. We define the **Reasoning Trace Gap** as the discrepancy between the

elicited, participant-endorsed checkpoint state and the supporting evidence in the frozen record, characterizing both the recovery need and the available material (Figure 1).

To explore an interaction approach to this need, we developed the **Reasoning Memory Workspace (RMW)**. RMW organizes access to a reconstructed reasoning position through a resume brief, reasoning cards, and a relationship network. These views provide an overview for reorientation, individual items for focused inspection, and relationships for examining how the reasoning fits together. RMW serves as a design probe for helping people re-enter prior reasoning during human-AI collaboration.

We address three research questions:

RQ1: Reasoning after interruption. Which components of participants’ endorsed Reasoning Positions remain available through unaided recall after interruption?

RQ2: Support from persistent records. To what extent does the frozen interaction record support these Reasoning Positions, and how does record support relate to unaided retention?

RQ3: Recovery support. How does immediate post-support restatement of the endorsed Reasoning Position compare across RMW, Self-authored Notes, and AI Summary?

We studied an LLM-assisted policy-analysis task with 33 participants. The platform froze each participant’s interaction record before T1, when participants articulated their endorsed Reasoning Position. After an interruption, unaided T2 assessed its retention. Across 252 reference propositions, 83 of 146 explicitly recorded propositions were not fully retained at T2, while 19 of 58 propositions with insufficient record support were fully retained. These complementary patterns identify content for which recovery support could help and context that a record alone cannot document.

After T2, participants reviewed RMW, Self-authored Notes, or AI Summary (11 per condition), followed by T3 and continuation with new evidence. RMW showed higher immediate checkpoint restatement after adjustment for T2 retention, providing initial evidence for this recovery approach. We interpret the findings as an exploratory evaluation of a concrete design direction, with the immediate recovery outcome defined by the repeated six-component prompts.

We make three contributions:

- (1) A six-component operationalization of Reasoning Position that makes the intended object of interruption recovery explicit and provides a shared participant-endorsed reference for evaluation.
- (2) An empirical characterization of incomplete post-interruption retention and its relationship to support in persistent records, identifying complementary resources and gaps for recovery.
- (3) RMW as an instantiation of reasoning-position-oriented recovery, with an exploratory comparison providing initial evidence for an interaction design direction that helps users reconnect with prior reasoning.

2 Related Work

2.1 Task Interruption and Resumption

Interruptions impose resumption costs, and those costs turn not only on whether people remember task-related facts but also on whether the interruption weakens the goals and intermediate representations that resumption requires. Memory-for-goals theory models a goal as an activation-based representation that decays over time and is vulnerable to interference [1, 2]. Problem State Theory adds that interruptions disrupt work because an intervening task can displace or overwrite the intermediate representations resumption depends on [4]. What happens before the interruption also matters: prospectively encoding what one intends to do next, and retrospectively reviewing how one arrived at the current stage, both shorten resumption time [46]. Studies of interrupted programming show that developers fall

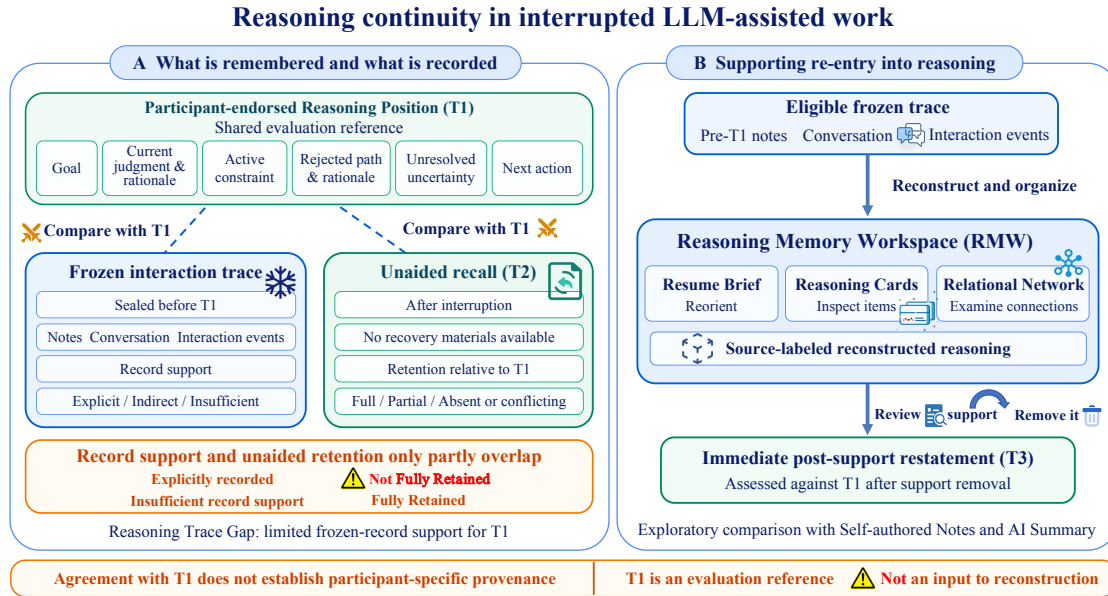


Fig. 1. Conceptual overview of the study. (A) Frozen interaction traces are examined against participants' endorsed Reasoning Positions at T1. (B) Recovery materials and a task-prior-only reconstruction are compared with the same reference, highlighting the distinction between checkpoint alignment and participant-specific provenance.

back on reconstructive strategies, and that cues restoring task *state* help more than cues that merely re-present task *content* [14, 33]. Work on source memory and prospective memory shows that remembering a fact, remembering where the fact came from, and remembering an intended action can fail independently of one another [11, 20].

Resuming a complex task therefore depends on more than recovering task content. Task goals give direction to current action [1, 2]; problem states hold intermediate judgments formed together with their associated constraints [33, 37]; retrospective information preserves prior reasoning processes and decision boundaries [31]; epistemic states separate what has been settled from what still needs verification [42, 44]; and prospective memory holds intended next actions [11, 46]. Building on these foundations, we use six components to specify a recovery target for LLM-assisted work. Prior work suggests that task information may be retained differently after interruption [16]. our study examines these component profiles together with the support available in persistent records, then explores how a recovery interface can make that reasoning accessible.

2.2 Interaction Traces and External Representations in Human-AI Collaboration

Theories of distributed and external cognition hold that problem solving is not confined to the individual but distributed across internal memory and external representations [39, 53], and that the form of an external representation shapes how people understand a task and how they proceed with it [32, 38]. In LLM-assisted knowledge work these representations are unusually rich: conversation histories, document revisions, material views and interface actions together form a digital trace of the task process, and preserving such fine-grained records can reveal how human-AI collaboration unfolds [24]. In principle, these traces might buffer the decay of internal memory.

However, activity traces are not the same as reasoning states. A document revision records what changed but not necessarily why. Viewing a source shows that a user inspected evidence but not whether the user accepted it [26, 52]; and a view that once appeared in a conversation may no longer be one the user endorses [30, 45]. For recovery design, what these records can contribute when users return matters. Comparing record support with unaided retention identifies where recorded reasoning can be made accessible and where recovery may require additional user context.

2.3 Constructing Structured States from LLM Conversations

A growing body of systems turns long-running projects into structured representations that support later revisitation, understanding and reuse. Sensemaking systems organize and compare large volumes of model-generated information [12, 43]; chaining systems expose intermediate steps [48, 49]; Memory Sandbox makes conversational memory visible and editable [18]; OnGoal tracks evolving conversational goals [7]; Branchat supports revisiting historical branches through a tree-structured interface [25]; ThinkFlow extracts cognitive workflows from conversations to support reusable human-AI common ground [5]. These systems provide precedents for making conversational goals, intermediate work, and relationships accessible. We build on this approach by organizing recovery around a participant’s reasoning position and assessing what the user can restate after reviewing the support. Model-side continuity research examines the information a system can retain and reuse [10, 27, 28]; task-resumption research highlights the user’s need to reactivate goals and recover prior task state [8, 46]. RMW brings these concerns together around re-entry into prior reasoning.

Interpreting a reconstructed position also requires care: plausible goals or next actions need not document what a particular user previously endorsed [3, 23]. We retain this distinction as a supplementary diagnostic of recovery materials, using a task-prior-only reconstruction to examine agreement that may arise without participant history [15, 19, 21]. Our contribution is a proposition-level account of how unaided recall and frozen-record support overlap and diverge during resumption. This motivates recovery designs that connect recorded reasoning with user-supplied context, which we begin to explore through RMW.

3 Reasoning Memory Workspace

Reasoning Memory Workspace (RMW) explores an interaction approach to reasoning continuity: helping a returning user locate a prior position, inspect its basis, and identify how to proceed. It organizes participant notes and conversation into a source-labeled reconstruction presented through coordinated views. The design makes the reasoning position an explicit object of recovery, while leaving the user responsible for assessing how that position relates to the task at hand.

3.1 Design Objective and Reasoning Position

The six components specify the functional aspects of reasoning that the workspace is intended to make accessible. They also provide the study’s framework for eliciting a participant-endorsed checkpoint reference: **Goal**, **Current Judgment and Rationale**, **Active Constraint**, **Rejected Path and Rationale**, **Unresolved Uncertainty**, and **Next Action**.

- **Goal**. The active decision objective or subgoal the user is pursuing, rather than the broader task topic [1, 5, 6].
- **Current Judgment and Rationale**. The user’s current position, together with its supporting reasons, qualifications, and conditions [35, 44, 51].
- **Active Constraint**. A requirement, threshold, or boundary condition that limits the acceptable options [4, 34].
- **Rejected Path and Rationale**. An alternative the user has removed from active consideration, together with the reasons for rejecting it [4, 34].

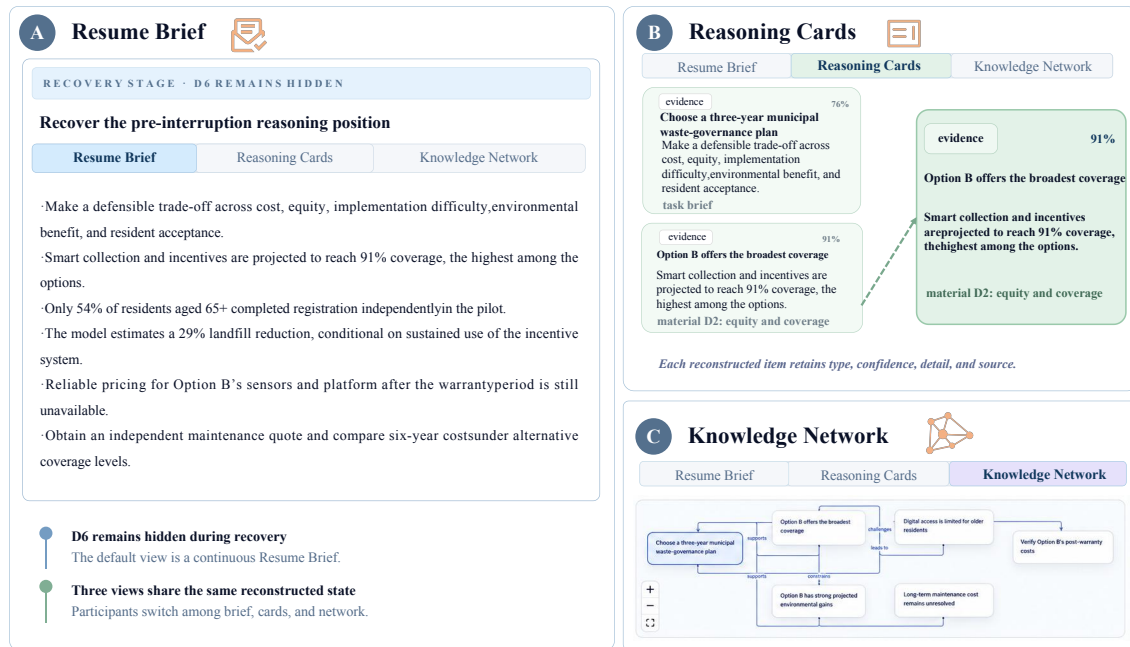


Fig. 2. Illustrative mockup of RMW's three recovery views: (A) Resume Brief, (B) Reasoning Cards, and (C) Knowledge Network. The views present the same reconstructed reasoning state through a brief, structured cards, and relationships among items. Cards include source labels and model-provided confidence. The example is illustrative; confidence values are not calibrated probabilities of correctness.

- **Unresolved Uncertainty.** A proposition the user regards as tentative, unverified, or dependent on further evidence [34, 35].
- **Next Action.** A specific step the user intends to take next, such as checking evidence, comparing options, or verifying an assumption [44, 51].

For example, remembering “I am leaning toward B” leaves open why B was favored and under what conditions. A position such as “I am leaning toward B, provided that maintenance responsibility is assigned and accessibility for older adults is validated” also identifies qualifications that matter for continuation. RMW's recovery target includes these qualifications, unresolved issues, and intended actions. The components are a task-specific operationalization. Their relationship to the platform schema and coding boundaries is illustrated in Appendix E.1.

3.2 Interface

RMW provides three levels of access to the reconstructed position (Figure 2). The design combines orientation, focused inspection, and relational navigation [38, 43, 46], allowing a user to begin with an overview and inspect particular reasoning items as needed.

- **Resume Brief:** The default view presents an overview of the reconstructed reasoning and intended next steps. Its purpose is to help users identify where their analysis had reached before deciding which details require attention.

- **Reasoning Cards:** Cards present a statement with its type, supporting detail, model-provided confidence, and a source label pointing to a memo, chat turn, task material, or user note. This format makes an individual judgment, condition, or unresolved issue available for inspection without rereading the full conversation. Source labels indicate where a claim could be checked; model-provided confidence is not a calibrated probability of correctness.
- **Knowledge Network:** Connected nodes expose labeled relations such as supports, challenges, constrains, rejects, and leads to. For a conditional recommendation, the user can inspect how evidence supports a judgment and how a constraint or unresolved issue bears on it. Relational access is intended to help reconnect pieces of reasoning whose significance depends on their connections.

These are complementary design intentions, evaluated here as a combined recovery interface. Participants assigned to RMW could move among the views after completing T2 and before indicating readiness for T3; D6 remained hidden throughout recovery. The system recorded view visits and the timing of user actions.

3.3 Experimental Recovery Conditions

The three conditions used the same task materials and interruption procedure. Recovery support differed in content and presentation:

- **Self-authored Notes:** participants revisit the notes they wrote before the interruption. Their content and organization remain as authored, without machine-generated structure.
- **AI Summary:** the interface presents a prose brief assembled from the participant’s extracted problem-state cards. The brief uses each card’s supporting detail, or its main statement when no detail is present.
- **RMW:** the interface presents a brief assembled by the same function, together with reasoning cards and a relationship network. Cards expose component type, goal level, confidence, and source labels. The network makes cross-component relationships navigable.

The implemented RMW and AI Summary interfaces use one extraction endpoint and one function for composing the prose brief. The intended difference is access to cards and the network. RMW–Notes additionally varies authored content and organization. We evaluate the delivered recovery conditions as packages.

4 Method

We conducted a study to characterize the recovery need and compare three forms of recovery support. The shared procedure through unaided T2 measured retention and its relationship to the frozen record. Recovery support appeared only after T2, enabling a between-subjects comparison of immediate post-support restatement (Figure 3).

4.1 Task Materials and Apparatus

Participants developed a recommendation among three municipal waste-management options using five evidence sets (D1–D5). The materials included verifiable facts, conditional constraints, conflicting information, and sources of varying reliability. No option dominated all criteria, so the task required evidence integration and reasoning under uncertainty. Prespecified answers let us distinguish factual accuracy from consistency with a participant’s reasoning (Appendix C).

The web interface provided an evidence viewer, a note editor, and a server-mediated DeepSeek assistant with a fixed, task-bounded prompt. The platform autosaved notes and conversations and logged material access, revisions, and stage transitions. A separate extraction step generated a structured problem state from the pre-checkpoint record. Participant-authored notes and chat could support participant-specific reasoning; assistant statements counted only

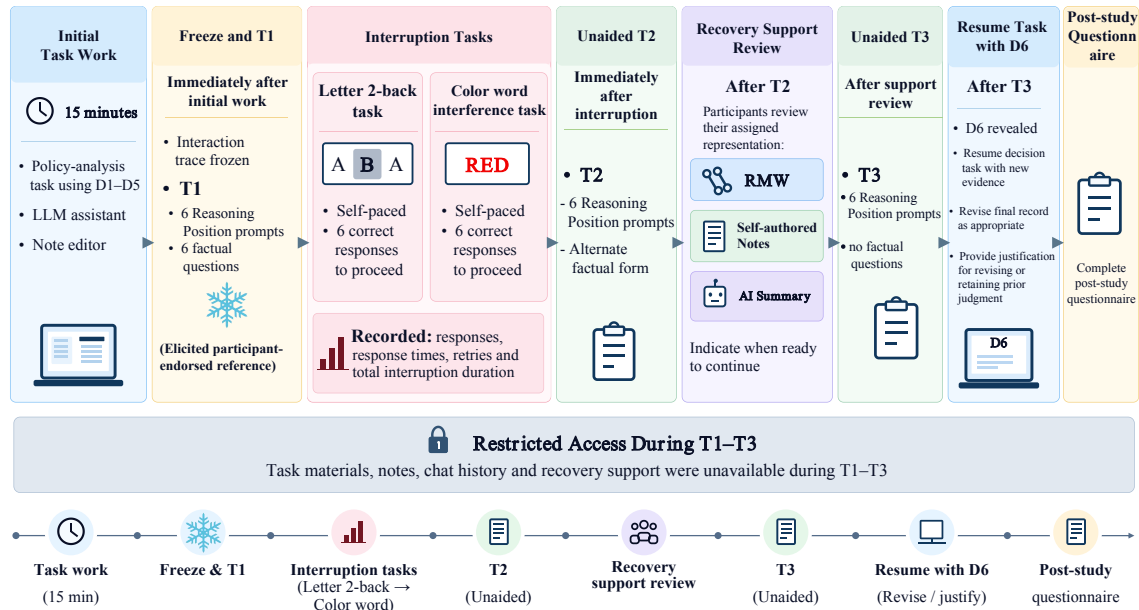


Fig. 3. Study procedure and information boundaries. The pre-checkpoint trace was frozen before T1. Recovery support was revealed only after unaided T2, and D6 was introduced after T3.

when the participant adopted or restated them. Appendix A specifies the implementation defaults, extraction schema, and missing runtime records.

Immediately before T1, the platform sealed the notes, conversation, viewed-material identifiers, and eligible events as an immutable frozen trace. T1–T3 responses, recovery interactions, and D6 were stored separately from the frozen trace. We assess the Reasoning Trace Gap by comparing the elicited T1 state with this earlier record. The analysis export contained 349 pre-T1 evidence records across the 33 participants: 155 material-access records, 118 note-revision excerpts, and 76 participant-chat excerpts.

4.2 Participants

We recruited 35 students enrolled in undergraduate or postgraduate programs. All participants provided informed consent and had no prior exposure to the study materials. Before the task, participants completed a 28-item questionnaire assessing generative AI use, AI literacy, research-task self-efficacy, prior topic exposure, and subjective prior knowledge.

After completing unaided T2, participants were randomly assigned by the study platform to one of three recovery conditions: RMW, AI Summary, or Self-authored Notes. The resulting allocation comprised 12 participants in RMW, 12 in AI Summary, and 11 in Self-authored Notes. Participants could neither select nor change their assigned condition. The platform then displayed the assigned recovery representation for review before T3. Two participants, one from RMW and one from AI Summary, were excluded because their T3 responses were incomplete, preventing calculation of the primary post-support retention outcome. The final analytic sample comprised 33 participants, with 11 per condition.

4.3 Procedure

Participants first worked for 15 minutes on the policy-analysis task using D1–D5, the LLM assistant and the note editor. Immediately afterward, the platform froze the interaction trace. Participants then completed T1, which established the participant-endorsed reference state using six Reasoning Position prompts together with six factual questions. T1 served as an elicited, participant-endorsed reference rather than a direct observation of the participant’s state before elicitation.

Participants next completed two self-paced interruption tasks in a fixed order: a letter 2-back task followed by a color-word interference task. Each task required six correct scored responses before participants could proceed, and the platform recorded responses, response times, retries, and total interruption duration. Immediately after the interruption, participants completed unaided T2 using the same six Reasoning Position prompts and the alternate factual form.

Recovery support was revealed only after T2. Participants reviewed their assigned representation, RMW, Self-authored Notes or AI Summary, and indicated when they were ready to continue. The support screen was then removed and participants completed T3, which repeated the six Reasoning Position prompts without factual questions. During T1–T3, task materials, notes, chat history and recovery support were unavailable from the assessment interface, and participants could not navigate back to them. Thus, T2 measured unaided retention after interruption, whereas T3 measured immediate restatement of the prior Reasoning Position after support review.

After T3, the platform revealed D6 and participants resumed the decision task with the new evidence. They revised their final record as appropriate, allowing us to assess integration of D6 and justification for either revising or retaining the prior judgment. A post-study questionnaire concluded the session.

4.4 Measures and Coding

4.4.1 Factual accuracy. T1 and T2 used alternate six-item factual forms derived from D1–D5. Participants were randomly assigned at the participant level to one of two form orders, AB or BA, with equal probability. Each response was scored as correct or incorrect against a form-specific answer key. Scoring accepted semantically equivalent wording and equivalent numerical or unit representations according to item-specific acceptance rules. Participant-level accuracy was the proportion of the six items answered correctly. We report the paired T1–T2 change together with sensitivity analyses by factual form and administration order. Appendix C summarizes the task materials, lists the Reasoning Position prompts, and provides the factual question targets and canonical answers.

4.4.2 Reasoning position retention. Each participant’s T1 responses were segmented into independently evaluable propositions assigned to the six components. The same propositions were matched against T2 and T3. Coders scored each as 0 (absent or conflicting), 1 (partially retained or materially altered), or 2 (essential content and epistemic status preserved). For participant i with M_i reference propositions, normalized retention was

$$R_{it} = \frac{1}{M_i} \sum_{j=1}^{M_i} \frac{r_{ijt}}{2}, \quad t \in \{T2, T3\}. \quad (1)$$

Scores range from zero to one and quantify consistency with the participant-endorsed T1 reference. We first average within each participant and then weight participants equally. Component estimates use the same procedure within each component. T1 equals one by construction. Reasoning loss is one minus T2 retention.

The final reference set contained 252 propositions across 33 participants (mean 7.64, SD 1.03; median 7; range 6–10). Of the 198 participant–component units, 144 (72.7%) contained one proposition and 54 (27.3%) contained two. This distribution describes the final shared coding set rather than independent segmentation agreement (Appendix E.3).

4.4.3 Frozen-trace support. For every T1 proposition, coders inspected the frozen trace and assigned 2 for explicit participant-specific support, 1 for indirect support requiring a defensible inference, or 0 for insufficient support. We report category counts and percentages, together with a normalized mean obtained by halving the codes and averaging within participant first. The normalized trace gap is one minus this mean; unlike the proportion of propositions with insufficient support, it also reflects partial gaps from indirectly supported propositions. Supporting spans or event identifiers link judgments to the record. A trace-support score locates evidence in the earlier record, while a representation-consistency score assesses the displayed recovery material. Insufficient trace support indicates that the frozen record does not adequately support an elicited T1 proposition. It does not distinguish previously unexpressed reasoning from content developed during T1 elicitation.

4.4.4 Coding procedure and reliability. Two coders established a shared proposition set and component assignments, calibrated the coding manual using practice examples, and independently scored all 252 propositions without access to each other’s judgments. Scores were linked to the relevant T2 or T3 responses, frozen-trace evidence, or displayed recovery materials. Disagreements were subsequently reviewed against the coding manual and source evidence to obtain the consensus codes used in the final analysis. Because both coders scored the same prespecified proposition set, agreement estimates reflect scoring reliability rather than independent segmentation reliability.

Coders were blinded to assigned condition during independent scoring. Condition identifiers were removed from the coding materials, and presentation-specific formatting that could reveal whether a record came from RMW, Self-authored Notes, or AI Summary was not shown to the coders. This masking was maintained for T2 and T3 retention, frozen-trace support, and representation-consistency scoring.

Pre-adjudication agreement was calculated from the two coders’ independent scores before any disagreements were discussed. Table 8 in Appendix E.2 reports the agreement statistics for each coded dimension. Ordinal 0–2 measures are summarized using linearly weighted Cohen’s κ , whereas binary measures use unweighted Cohen’s κ . Consensus scores obtained after adjudication were used for all subsequent analyses.

4.4.5 Post-support change and continuation. Immediate post-support alignment was measured as T3 retention relative to the participant-endorsed T1 Reasoning Position, using the same proposition-level scoring procedure described above. T3 assessed participants’ ability to restate that reference immediately after reviewing the assigned support, with the support no longer visible. The primary condition comparison evaluated T3 retention while adjusting for participants’ unaided T2 retention, with T3–T2 gain reported descriptively. After T3, participants received D6 and resumed the decision task; evidence integration and justified retention or revision of the prior judgment were coded separately as secondary continuation outcomes.

4.5 Supplementary Recovery-Material Diagnostics

To contextualize the recovery comparison, we assessed the consistency of selected displayed materials with T1 and separately scored a reconstruction generated without participant history. These diagnostics distinguish agreement with the reference from support in the participant’s record. They are supplementary to RQ1–RQ3; their scoring procedure and full displayed-material summaries appear in Appendix G.

4.6 Analysis

All analyses were exploratory; the study was not preregistered. The dataset contains 252 propositions, 396 factual responses and 33 post-task questionnaires. Reasoning estimates average within participant first; component estimates do so within component. Unless specified otherwise, 95% confidence intervals use a participant-level t distribution. Proposition counts and support percentages use a different denominator from these equally weighted participant means.

- **RQ1: Reasoning after interruption.** We summarize unaided T2 Reasoning Position retention relative to T1, overall and across the six components. Factual-accuracy change provides additional context and is assessed using paired summaries and an item-level binomial generalized estimating equation.
- **RQ2: Support from persistent records.** We summarize frozen-trace support overall and by component, then cross-classify support with unaided T2 retention on the same T1 propositions. Associations among component-level profiles are supplementary descriptive analyses.
- **RQ3: Recovery conditions.** We model participant-level T3 retention as a function of T2 retention and recovery condition, with RMW as the reference condition. The two focal contrasts compare RMW with Self-authored Notes and AI Summary. We report adjusted differences with 95% confidence intervals and multiplicity-adjusted p -values. T3–T2 gain is reported descriptively.

Supplementary analyses characterize displayed-material and task-prior-only consistency (Appendix G) and examine sensitivity of component profiles to exported reference-text length (Appendix F.2). The latter addresses a limited proxy for response complexity.

5 Results

Results include 33 participants (11 per condition) and 252 T1 reference propositions. We first characterize which parts of the reasoning position remain available after interruption, then examine support in the persistent record, and finally compare recovery outcomes across conditions.

5.1 Availability of the Reasoning Position after Interruption (RQ1)

Unaided Reasoning Position retention averaged 0.655 (SD 0.183, 95% CI [0.590, 0.720]) relative to the participant-endorsed T1 reference. Retention varied descriptively across components (Table 1). **Current Judgment and Rationale** had the lowest mean (0.515), followed by **Unresolved Uncertainty** (0.553) and **Next Action** (0.614). **Goal** (0.818) and **Rejected Path and Rationale** (0.788) had the highest means. Thus, retaining the task goal coexisted with less complete retention of the current judgment, its qualifications and unresolved or prospective aspects of the task.

As additional task context, factual accuracy declined from 80.3% (SD 16.9%) at T1 to 63.6% (SD 17.9%) at T2, a paired mean decline of 16.7 percentage points (95% CI [7.9, 25.4]). The item-adjusted GEE estimated lower odds of a correct response at T2 (OR 0.411, 95% CI [0.260, 0.651], $p < .001$). Mean accuracy declined in both factual-form orders (Appendix F.2).

The descriptive component ordering persisted after adjustment for exported reference-text length; this sensitivity check does not account for the full complexity of the original responses (Appendix F.2).

5.2 Persistent Records as a Resource for Recovery (RQ2)

5.2.1 Trace coverage differed across components. Of the 252 propositions, 146 (57.9%) had explicit participant-specific support in the frozen trace, 48 (19.0%) had indirect support, and 58 (23.0%) had insufficient support. Participant-level

Table 1. Unaided T2 retention, frozen-trace support, and task-prior-only consistency by reasoning component. Retention and prior-only means average within participant first ($n = 33$ in every row); support columns are percentages of propositions, and m is the proposition count. Overall means weight participants equally over their own proposition sets.

Component	m	T2	95% CI	Explicit	Indirect	Insuff.	Prior
Goal	35	0.818	[0.723, 0.913]	82.9	8.6	8.6	0.811
Current Judgment and Rationale	49	0.515	[0.402, 0.628]	71.4	16.3	12.2	0.545
Active Constraint	45	0.697	[0.582, 0.812]	57.8	15.6	26.7	0.417
Rejected Path and Rationale	42	0.788	[0.681, 0.894]	54.8	31.0	14.3	0.280
Unresolved Uncertainty	42	0.553	[0.457, 0.649]	45.2	14.3	40.5	0.227
Next Action	39	0.614	[0.517, 0.710]	35.9	28.2	35.9	0.235
Overall	252	0.655	[0.590, 0.720]	57.9	19.0	23.0	0.414

Table 2. Frozen-trace support crossed with unaided T2 retention on the same 252 propositions. Entries are proposition counts, not independent participant observations. Retention indicates consistency with the T1 reference.

Trace support	Absent/conflicting (0)	Partial (1)	Full (2)	Total
Explicit	16	67	63	146
Indirect	5	19	24	48
Insufficient	9	30	19	58
Total	30	116	106	252

normalized trace support averaged 0.683 (95% CI [0.626, 0.741]), corresponding to a normalized trace gap of 0.317. **Goal** had the highest explicit coverage (29/35, 82.9%), followed by **Current Judgment and Rationale** (35/49, 71.4%). **Next Action** had the lowest explicit coverage (14/39, 35.9%), while **Unresolved Uncertainty** most often lacked sufficient support (17/42, 40.5%; Figure 4A).

5.2.2 Explicit recording did not ensure full unaided retention. Cross-classifying the same propositions revealed mismatches between recording and retention (Table 2). Of the 146 explicitly recorded propositions, 67 were partially retained and 16 were absent or conflicting at T2. Together, these 83 propositions (56.8%) occurred across 31 participants. Conversely, 19 of the 58 propositions with insufficient trace support (32.8%) were fully retained, across 14 participants.

The component profiles identify where these mismatches were concentrated (Figure 4). **Current Judgment and Rationale** combined the lowest mean retention with the second-highest explicit coverage: 29 of its 35 explicitly recorded propositions were not fully retained. **Unresolved Uncertainty** and **Next Action** were below the overall means for both retention and explicit coverage. **Rejected Path and Rationale** showed a different profile, with relatively high retention despite explicit coverage below the overall mean. These patterns distinguish recorded reasoning that was incompletely reproduced from reasoning that remained available despite insufficient support in the frozen record.

5.2.3 Supplementary context: recovery-material consistency. Selected-material consistency with T1 averaged 0.695 for RMW cards, 0.539 for Self-authored Notes, and 0.704 for AI Summary; these scores cover different material scopes and are not measures of users' recovery. A task-prior-only reconstruction also achieved agreement (0.414), illustrating why agreement alone does not document participant-specific reasoning. Full summaries appear in Appendix G.

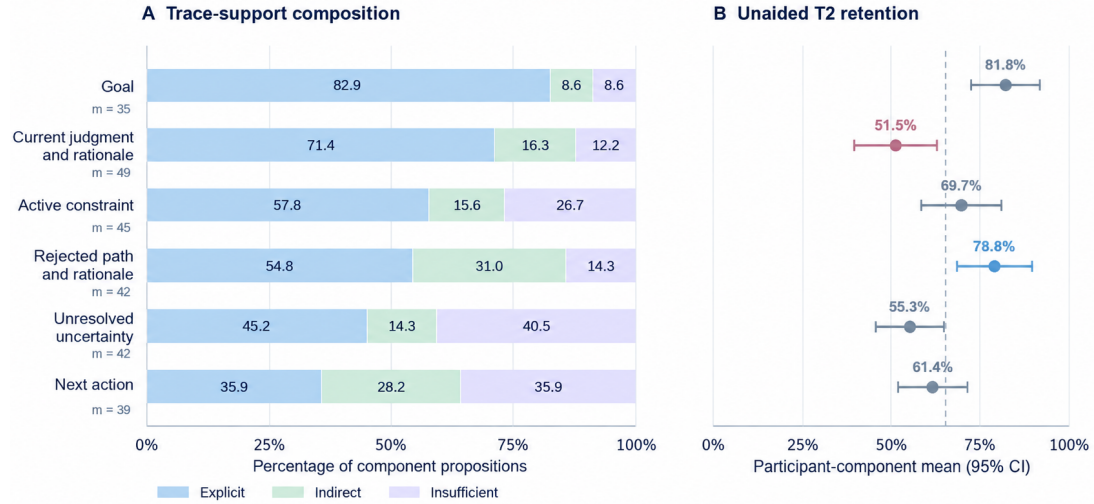


Fig. 4. Frozen-trace coverage and unaided reasoning retention by component. (A) Stacked bars show the percentages of component propositions with explicit, indirect, and insufficient support; m gives the proposition count. (B) Points and intervals show participant-first mean T2 retention and 95% confidence intervals ($n = 33$ per component). The dashed line marks overall T2 retention (0.655).

Table 3. Participant-level retention and change ($n = 11$ per condition). Entries are mean (SD). Gain is T3 minus T2 retention, calculated within participant before averaging. T3 measures immediate consistency with T1 after support review.

Condition	T2	T3	Gain
RMW	0.661 (0.169)	0.905 (0.066)	0.244 (0.152)
Self-authored Notes	0.647 (0.247)	0.797 (0.172)	0.149 (0.109)
AI Summary	0.657 (0.133)	0.817 (0.102)	0.160 (0.104)

5.3 Immediate Recovery across Support Conditions (RQ3)

RMW showed the highest mean immediate post-support retention (0.905), followed by AI Summary (0.817) and Self-authored Notes (0.797). Unaided T2 means were descriptively similar across conditions (0.647–0.661). Within-participant gains averaged 0.244 for RMW, 0.160 for AI Summary, and 0.149 for Self-authored Notes (Table 3; Figure 5). Across all participants, retention was 0.655 at T2 and 0.840 at T3, with a mean gain of 0.184 (95% CI [0.135, 0.234]). These gains describe change after support review; the condition comparison below addresses the relative outcomes of the three recovery approaches.

After adjustment for T2 retention, RMW exceeded Self-authored Notes by 0.102 (simultaneous 95% CI [0.017, 0.186], $p_{\text{adj}} = .016$) and AI Summary by 0.086 (simultaneous 95% CI [0.0006, 0.1713], $p_{\text{adj}} = .048$). Both adjusted differences remained positive in leave-one-participant-out refits, but the comparison with AI Summary was more sensitive to individual observations (Appendix F.2). These results provide initial evidence for RMW's immediate recovery value within the delivered conditions.

After T3, participants resumed the task using the new D6 evidence. Evidence integration was observed for 6/11 participants in the RMW condition, compared with 5/11 in both the Self-authored Notes and AI Summary conditions; justified retention or revision of the recommendation was observed for 5/11, 3/11, and 4/11 participants, respectively.

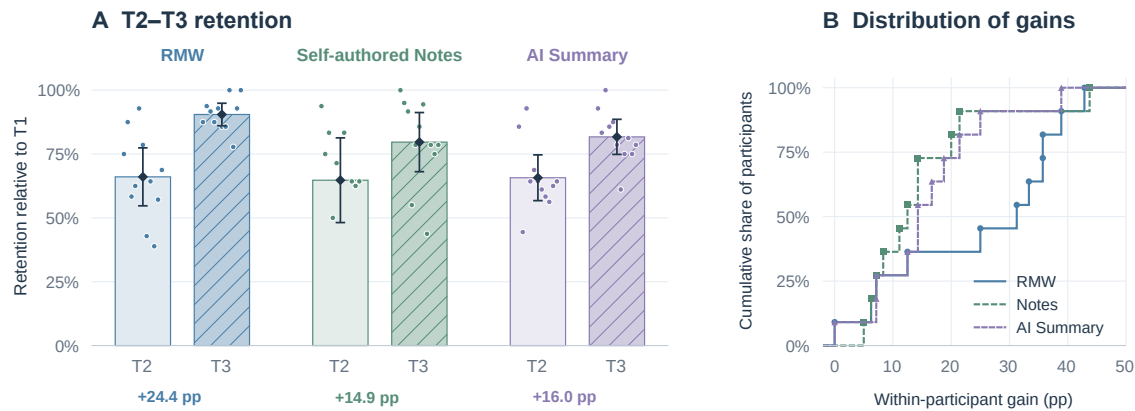


Fig. 5. Immediate post-support recovery by condition ($n = 11$ per condition). (A) Bars show unadjusted mean retention relative to T1 at unaided T2 and immediate post-support T3; dots show participant-level scores. Error bars are pointwise 95% confidence intervals for each mean, computed across participants using a t distribution with 10 degrees of freedom. Labels below show mean within-participant T3–T2 gains in percentage points (pp). (B) Empirical cumulative distributions show the proportion of participants whose T3–T2 gain is at or below each value. Notes denotes Self-authored Notes.

Although RMW was descriptively highest on both continuation outcomes, these small counts do not establish a reliable condition advantage. Nevertheless, together with the immediate recovery results, this pattern provides encouraging preliminary evidence for RMW as a design direction for supporting continued reasoning.

6 Discussion

The study motivates an interaction design direction for human-AI collaboration centered on helping users re-enter their reasoning position. The interruption task makes this need concrete. Users could retain the broad goal while incompletely reproducing the judgment, qualifications, uncertainties, and intentions that shaped their analysis. RMW provides one way to organize access to this reasoning, and its immediate recovery results offer initial support for developing the approach further.

6.1 Reasoning Position as an Object of Recovery

Goal retention was relatively high, while **Current Judgment and Rationale**, **Unresolved Uncertainty**, and **Next Action** were less completely retained. This pattern connects goal reactivation [1] with the intermediate representations emphasized by problem-state accounts of interruption [4]. Related work further suggests that variation in resumption costs may reflect the need to reconstruct the pre-interruption task context [17]. Together, these perspectives specify an interaction target: a returning user needs access to the position they had reached, including what made it defensible and what remained open.

The six components provide a practical vocabulary for this target. In the task, recovering a preference for B could leave its maintenance conditions or intended verification step unavailable. A recovery interface can make these aspects accessible together, helping the user inspect a judgment in the context that gave it meaning. This design direction aligns with proposals to support planning, self-evaluation, and self-management in generative AI interaction [44]. It also relates to ThinkFlow’s use of structured cognitive workflows to support intent expression and reuse across sense-making tasks [5], while our focus is on recovering a participant’s reasoning position after interruption. The component ordering is descriptive and task-specific: differences in qualification burden and the requirements for full credit may contribute to the observed retention differences.

6.2 RMW as a Direction for Human-AI Interaction

RMW translates this recovery target into a concrete interaction arrangement. A brief provides an entry point, cards expose reasoning items for inspection, and a network makes their relationships navigable. The design offers a way to move from a broad orientation to the particular condition, uncertainty, or dependency a user needs to examine [41]. This arrangement connects with recent work on reviewing evolving conversational goals and supporting selective inspection of LLM processes [7, 32]. RMW focuses these design concerns on helping a returning user recover access to the reasoning position from which their work can continue [47].

More broadly, RMW suggests a design direction for human-centered reasoning memory. In work that unfolds across multiple sessions, access to prior information may coexist with difficulty recovering why a judgment was reached, which conditions supported it, and what remained to be done [47, 55]. Model-side continuity presents a related but distinct challenge: research on multi-turn interaction shows that LLMs can remain anchored to early assumptions and fail to recover from an incorrect conversational trajectory [22]. These challenges motivate extending the goals of conversational memory beyond model-side recall and personalization to include people’s ability to recover, inspect, question, and revise their own reasoning [45]. In this direction, memory becomes an explicit representation of a developing reasoning position that users can examine and use to guide subsequent collaboration. Recovery includes reconsidering whether an earlier judgment still holds and identifying what should change before proceeding [50].

The RQ3 comparison provides initial evidence for this direction (Section 5.3). RMW participants showed higher immediate restatement than those receiving Self-authored Notes or AI Summary after adjustment for unaided T2 retention. This finding supports the feasibility of helping users regain access to a prior reasoning position through an organized recovery representation. The contribution is an implemented approach and an exploratory evaluation that make human-centered reasoning memory concrete as an interaction design direction. Future work should examine whether allowing users to correct and update this representation helps both users and models work from an explicitly reviewed account of the task, and whether this improves reasoning quality and continuity across longer periods of collaboration.

6.3 Combining Recorded Reasoning with User Context

The memory-record comparison highlights complementary roles for recorded reasoning and user context in recovery. Most explicitly recorded propositions that were not fully retained were partially retained, suggesting that recovery may involve restoring a qualification or dependency around a remembered claim. Recorded material can therefore provide useful context even when users recall the gist. This role relates to memory-augmentation systems that retrieve contextually relevant information to support users’ ongoing activities [56]. PaperTrail offers a related design precedent by connecting claims in LLM-generated answers to source evidence and making gaps in support available for inspection [30].

For recovery design, this suggests presenting reconstructed reasoning items alongside their recorded basis, so that users can inspect the conditions and evidence associated with a remembered conclusion.

Conversely, users fully retained some propositions with insufficient frozen-record support, making their recollections a resource for checking and supplementing a reconstruction. In a different context, research on LLM-assisted recall of witnessed events found that users drew on what they remembered and asked GPT to help identify omitted details, while their interactions could also shape how they interpreted those events [54]. This motivates treating users as active contributors to recovery while distinguishing their recollections from system-generated suggestions. Opportunities to clarify and correct reconstructed content also connect with work on revisable memory and feedback for maintaining human-AI common ground [45]. The supplementary task-prior diagnostic reinforces the importance of this distinction: a statement that matches the participant-endorsed checkpoint may arise from shared task information without documenting that participant’s earlier reasoning. Recovery interfaces should therefore make recorded sources accessible and allow users to articulate missing context while keeping the origins of these contributions visible.

The comparatively limited retention and explicit coverage of **Unresolved Uncertainty** and **Next Action** suggest a specific opportunity to connect recovery with lightweight capture. At a natural stopping point, a system could invite users to articulate an unresolved issue and their intended next action, retaining both as editable participant statements. This connects prospective preparation with retrospective review [46] and provides a way for users to supply context that the existing record may not contain.

7 Limitations and Future Work

This exploratory study evaluates one policy-analysis task with a student sample and 11 retained participants per condition. The estimates, particularly RMW-AI Summary, should be read alongside the sensitivity analyses (Appendix F.2). The participant-endorsed T1 reference may have been rehearsed or reshaped through elicitation, and insufficient frozen-trace support cannot distinguish previously unexpressed reasoning from content developed at T1. Component scores also differ in content and qualification demands, so their ordering is task-specific. Without uninterrupted and no-support controls, within-person changes cannot isolate interruption or support effects from elapsed time and repeated assessment. Replication should vary tasks, expertise, and control conditions.

T3 measures immediate consistency with T1 after support removal under repeated six-component prompts; correspondence between those prompts and recovery content may contribute to performance. The small descriptive D6 counts cannot determine condition differences in continuation. Longer delays and independent assessments of reasoning with new evidence are therefore needed to establish whether immediate restatement translates into sustained understanding or better decisions. Such evaluations should also test the lightweight capture proposed in Section 6.3, assessing whether recording unresolved issues and intended actions improves delayed recovery enough to justify the capture effort.

The comparison evaluates the recovery conditions as delivered. RMW-Notes varies content and presentation; although RMW and AI Summary share an extraction endpoint and a prose-composition function, missing per-session extraction requests and configuration records prevent verification of identical generation settings and input delivery. The study therefore cannot isolate the contributions of cards and relational views, and the task-prior diagnostic is not a matched ablation of participant history. Future comparisons should match underlying content, document generation inputs and settings, and assess which views improve access relative to their inspection cost. These comparisons would clarify how capture and presentation can support recovery, reassessment, and continuation across sessions.

8 Conclusion

Resuming LLM-assisted work involves recovering where one’s reasoning stood, including the basis and conditions of a judgment, unresolved questions, and intended next steps. We operationalized this recovery target as Reasoning Position and observed incomplete unaided retention alongside uneven support in persistent records. RMW instantiates an interaction design direction that makes this position accessible through coordinated views. Its exploratory advantage in immediate post-support restatement provides initial support for developing interfaces that help people reconnect with prior reasoning during human-AI collaboration. Establishing benefits for sustained reasoning and subsequent task performance is the next step in evaluating this direction.

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B Implementation and Evidence Boundaries

All formal-study sessions used the same OpenAI-compatible DeepSeek Chat Completions endpoint and the same deployed generation configuration. The model was deepseek-v4-flash for all participants. Chat requests disabled reasoning, used temperature 0.3, and limited output to 900 tokens. Extraction requests likewise disabled reasoning and used JSON-object mode, temperature 0, and a 3,200-token limit.

B Chat and State-Extraction Prompts

The following boxes present abridged versions of the implemented prompt templates. Bracketed placeholders denote dynamically inserted inputs. The chat prompt was localized to the interface language; the extraction prompt specified the output language.

B.1 Chat Prompt

During the initial work phase, the evidence pack contained D1–D5. During the continuation phase, it additionally contained D6. Chat generation used temperature = 0.3, max_tokens = 900, and thinking.type = disabled.

B.2 State-Extraction Prompt

The extraction schema accepts 2–12 cards and up to 20 relations. Cards contain identifier, kind, content, detail, status, priority, confidence, source, and justification fields; goals may also have a goal level. Card kinds are goal, hypothesis, evidence, constraint, path, and next action. Relations use supports, challenges, constrains, rejects, or leads-to labels. These types do not map one-to-one onto the measurement components: uncertainty and rejected paths may be expressed through content and status (Appendix E.1). The prompt directs the extractor to omit unsupported categories, treat assistant turns as context unless explicitly adopted, and use actions as progress signals rather than evidence of belief. Model-provided confidence is not an empirically calibrated probability.

The frozen record contains the pre-T1 memo, conversation, material identifiers, and eligible action sequence; T1–T3 and D6 are excluded. The extraction request contains the task identifier, language, memo, conversation, and actions. The

Table 4. The abridged chat prompt used during the research task. Italicized placeholders indicate session-specific inputs.

Chat Prompt Template**System message (abridged)**

You are the AI assistant in a research task. The student is investigating: *[Task question]*.

Talk like a natural, patient research partner. Respond to the student's current question in light of the preceding conversation. Answer direct questions directly, work through open questions collaboratively, and ask a brief clarifying question when needed. Do not repeatedly provide a complete analysis of the evidence pack.

Do not automatically endorse the student's proposal. When the student asks whether a causal claim is correct, first ask one or two specific questions that help them examine the supporting evidence, alternative explanations, confounds, or counterexamples. Provide your judgment when the student explicitly requests a direct conclusion or has already worked through the key trade-off. Do not assume that a before-and-after comparison controls for time trends, concurrent events, measurement changes, or observation effects.

Normally use one to three short paragraphs. Use lists only when helpful or requested, and avoid imposing a fixed response structure.

Cite material identifiers, such as [Material D1], for concrete facts or consequential claims. Clearly distinguish inferences and uncertainty. Do not introduce facts outside the evidence pack, assume a single correct framing, or write the student's complete final memo.

Evidence pack: *[Materials available in the current phase]*.

User and assistant messages

[Conversation history, preserving message roles and order, and ending with the participant's current message]

Table 5. The abridged state-extraction prompt and user message template. Italicized placeholders indicate session-specific inputs.

State-Extraction Prompt Template**System message (abridged)**

Extract the participant's prospective reasoning state immediately before an interruption. Return only a JSON object containing cards and relations. Write card text in *[Output language]*.

Treat the participant-authored memo and user chat turns as the only evidence of reasoning content. Assistant turns provide context only: include an assistant suggestion only if a later user turn or the memo explicitly adopts it. Failure to reject a suggestion does not constitute adoption.

Use the action trace only to infer attention, progress, sequence, and the last active step. Opening a material or checking a criterion does not indicate agreement. Use the task question only to contextualize the main goal. Do not import reasoning content from task materials alone, infer a hidden answer key, or identify a framing as correct.

Include exactly one main goal. Add only supported subgoals, hypotheses, evidence, constraints, suspended goals, rejected paths, and next actions. Aim for 4–10 concise, non-duplicative cards when supported, with no more than 12 cards. Omit unsupported categories rather than creating placeholders.

Each card must contain id, kind, content, detail, status, priority, confidence, source, and why. Card kinds are goal, hypothesis, evidence, constraint, path, and next_action. Use goalLevel only for goals, with values main, subgoal, or suspended.

Mark a rejected path as expired only when explicitly rejected by the participant. An inferred next action must be marked uncertain, with confidence no greater than 40 on a 0–100 scale. Cite identifiable sources without fabricating locations. Pin the main goal and next action; default other cards to normal priority. Treat all extracted cards as candidates requiring participant calibration.

Construct relations only between extracted cards and only when supported by the participant trace. Relation types are supports, challenges, constrains, rejects, and leads_to. Each relation must include an identifier, source and target card identifiers, a relation type, and confidence. Inferred relations must have confidence no greater than 40. Direct leads_to from an earlier reasoning state or path toward a next action, not in the reverse direction. An empty relation list is valid.

Limit card content to 32 Chinese characters or 18 English words, and detail to 70 Chinese characters or 45 English words.

User message template

Task question: *[Task question]*

Participant memo: *[Participant's memo]*

Conversation: *[Numbered user and assistant turns]*

Research action trace (progress signals only, not evidence of belief): *[Ordered action records]*

server assembles model input from the task question, memo, conversation, and actions without separately appending the complete D1–D5 documents.

AI Summary and RMW share a function that concatenates card details, falling back to the main statement when detail is empty. This brief requires no additional LLM request; RMW additionally renders cards and relations. Displayed-material analysis evaluates the selected material from each participant’s assigned condition. The task-prior-only diagnostic is assessed separately against T1; the comparison is not a matched ablation of access to participant history.

C Task Materials and Assessment Prompts

The task asks which of three waste-governance options a municipality should adopt over three years: A adds community staff and mobile collection; B uses smart collection points with incentives; C builds centralized mechanical sorting. Participants weigh cost, equity, implementation, environmental benefit, and resident acceptance. The following summaries reflect the current Chinese stimuli in the supplied platform source.

D1: Costs. The three-year ceiling is RMB 180 million. Estimated costs are RMB 132 million (A), 165 million (B), and 176 million (C), including construction and three years of operation but excluding later equipment renewal.

D2: Equity. A offers face-to-face help but projected suburban coverage is 62%. B offers 91% coverage but requires smartphone registration; 54% of residents over 65 registered independently in the pilot. C has more even coverage but does not address dirty public spaces at the collection stage.

D3: Implementation and acceptance. Pilot satisfaction is 78% (A), 69% (B), and 74% (C). Staff turnover for A is 31%; 8% of B collection records were incorrectly credited. Opposition near C sites is 46%, with additional environmental assessment requested. Pilot populations are not fully comparable.

D4: Environmental model. Projected landfill reductions are 18% (A), 29% (B), and 34% (C), conditional on design throughput. Transport offsets approximately seven percentage points of C’s emissions benefit. B depends on continued incentive use; A’s parameters are closest to existing local data.

D5: Long-term operation. Agencies favor different options. B lacks reliable maintenance quotes after year three, while C may face a 12–18-month approval delay. No option dominates all five criteria.

D6: New evidence. An unaudited supplier quote shortens B’s sensor warranty from five to three years and estimates annual maintenance in years four to six at 2.4 times the earlier estimate. Keeping all points operating puts six-year cost 22% above C; reducing maintained points lowers suburban coverage from 91% to 68%. D6 appears only after T3.

Table 6. Reasoning Position prompts, using the platform’s English wording.

Component	Prompt
Goal	What core goal were you pursuing before the interruption?
Current Judgment and Rationale	Where had your reasoning reached? State the current position or design and its key basis.
Active Constraint	What key constraint was still binding your reasoning?
Rejected Path and Rationale	What direction had you rejected, and why?
Unresolved Uncertainty	Which unresolved uncertainty was most likely to change the current judgment?
Next Action	What was the minimum next action you intended to take?

D Post-Task Instrument

The post-task questionnaire contained 14 Chinese-language items rated on a seven-point agreement scale (1 = strongly disagree; 7 = strongly agree), covering task load (five items), perceived loss (three), metacognition (two), AI reliance

Table 7. Factual forms and answer keys represented in the study materials.

Form	Item	Question target and answer
A	A1	Three-year budget ceiling: RMB 180 million.
	A2	A's three-year cost: RMB 132 million.
	A3	B's projected coverage: 91%.
	A4	C's possible approval delay: 12–18 months.
	A5	A's resident satisfaction: 78%.
	A6	B's modeled landfill reduction: 29%.
B	B1	C's three-year cost: RMB 176 million.
	B2	A's suburban coverage: 62%.
	B3	Independent B registration among residents over 65: 54%.
	B4	B trial records not correctly credited: 8%.
	B5	Opposition around C candidate sites: 46%.
	B6	C's benefit offset by transport: 7 percentage points.

(two), and agency (two). All 33 retained participants provided complete responses. These supplementary self-reports are not primary outcomes for the research questions; their composites are not treated as validated latent scales.

E Coding Manual and Reliability

The analysis workbook implements the following final-code definitions. Reasoning retention is coded 0 when the T2/T3 response is absent from or conflicts with the participant's T1 proposition, 1 when the gist remains but a material rationale, qualifier, condition, or epistemic status is lost, and 2 when essential content and epistemic status are substantially preserved. Trace support is coded 0 for no participant-specific pre-T1 evidence, 1 for indirect support requiring one defensible inference, and 2 for explicit participant-specific support. Task plausibility or a material-view event alone does not count as participant-specific support.

Representation consistency uses the same 0–2 logic against T1: 0 for missing or conflicting content, 1 for gist-level agreement with altered detail or certainty, and 2 for substantial agreement in both essential content and status. A pointer is valid only if it resolves uniquely to an eligible frozen participant-authored or explicitly adopted span. Unsupported addition and certainty inflation are coded independently as binary error flags. D6 evidence integration requires the final memo to connect D6 to a prior constraint or judgment; justified revision requires a reasoned explanation for preserving or changing the prior recommendation.

E.1 Worked Component-Boundary Example

The platform schema and the measurement dimensions are related but not identical. The schema provides *hypothesis* and *constraint* cards, together with an *uncertain* status, whereas the measurement rubric assigns T1 propositions to six dimensions according to their function in the participant's reasoning. The following example shows how we applied this distinction in the final coding.

Anonymized coding example (translated from Chinese).

Condensed T1 excerpts. “I conditionally favor B because it offers 91% coverage and a projected 29% reduction in landfill. I would continue to support B only if barriers for older residents were reduced and responsibility for maintenance after year three were clarified. Smartphone registration is a binding equity concern because only 54% of older residents

registered independently. Reliable post-warranty maintenance costs and the responsible department remain unresolved and could change my judgment.”

Current Judgment and Rationale. The conditional preference for B, its supporting evidence, and the conditions attached to continued endorsement.

Active Constraint. The smartphone-registration requirement and the 54% independent-registration rate, because they already restrict whether B meets the participant’s equity criterion.

Unresolved Uncertainty. The unavailable maintenance price and unclear responsibility after year three, because resolving them could change the participant’s judgment.

Segmentation and deduplication rule. Coders classified propositions by functional role rather than by topic. A clause beginning “only if” stayed with the Current Judgment when it qualified endorsement; an operative boundary was coded as an Active Constraint; and missing evidence capable of changing the judgment was coded as an Unresolved Uncertainty. Repeated wording with the same content and role was counted once. A topic could occur in more than one dimension only when distinct propositions gave it different functions—for example, maintenance responsibility as a condition on endorsement and missing maintenance evidence as an unresolved uncertainty.

E.2 Independent Coding, Blinding, and Agreement

Table 8 reports pre-adjudication agreement from the two coders’ independent judgments. Exact agreement is the percentage of identical ratings. Ordinal 0–2 dimensions use linearly weighted Cohen’s κ ; binary dimensions use unweighted Cohen’s κ . Analyses use the adjudicated consensus scores. These estimates concern scoring on the shared proposition set, not independent segmentation. The independent scoring and condition-masking procedures are described in Section 4.4.4.

Table 8. Pre-adjudication coding reliability. Agreement denotes exact agreement between coders; κ denotes Cohen’s kappa.

Dimension	Unit	n	Agreement	κ	95% CI
T2 retention	Proposition	252	88.5%	0.82	[0.77, 0.87]
T3 retention	Proposition	252	90.1%	0.85	[0.80, 0.89]
Trace support	Proposition	252	86.9%	0.80	[0.74, 0.85]
Displayed consistency	Proposition	252	84.5%	0.76	[0.70, 0.82]
Task-prior consistency	Proposition	252	83.7%	0.74	[0.68, 0.80]
Unsupported addition	Proposition	252	93.3%	0.78	[0.68, 0.87]
Certainty inflation	Proposition	252	92.5%	0.75	[0.64, 0.85]
D6 evidence integration	Participant	33	87.9%	0.76	[0.55, 0.92]
D6 justified revision	Participant	33	84.8%	0.70	[0.47, 0.88]

E.3 Distribution of the Final Reference Propositions

The final coding set contained 6, 7, 8, 9, and 10 propositions for 3, 14, 10, 4, and 2 participants, respectively. Each participant contributed at least one proposition to every component. Table 9 summarizes the final assignments. These counts do not independently validate the original segmentation boundaries or component assignments.

Table 9. Distribution of final reference propositions across participant–component units. Each row contains 33 participants; no unit contained zero or more than two propositions.

Component	One	Two	Propositions
Goal	31	2	35
Current Judgment and Rationale	17	16	49
Active Constraint	21	12	45
Rejected Path and Rationale	24	9	42
Unresolved Uncertainty	24	9	42
Next Action	27	6	39
Total	144	54	252

F Additional Analysis Details

F.1 RQ3 Model and Inference

The primary condition model uses normalized participant-level retention as defined in the main text:

$$R_{i,T3} = \beta_0 + \beta_1 R_{i,T2} + \beta_2 \mathbb{I}[\text{Notes}_i] + \beta_3 \mathbb{I}[\text{AI Summary}_i] + \epsilon_i. \quad (2)$$

Here $\mathbb{I}[\cdot]$ is an indicator and RMW is the reference condition. The focal contrasts are $-\beta_2$ (RMW minus Notes) and $-\beta_3$ (RMW minus AI Summary).

The model contains four parameters and 33 participants. The two focal contrast intervals use $t_{29,.9875}$ with HC3 standard errors, applying a Bonferroni correction for the two-contrast family. Derived T2 and T3 outcomes were recomputed from the final proposition-level codes; all 33 participant means agree with the participant–outcome values.

F.2 Sensitivity Analyses

Factual forms. Eighteen participants received order AB and 15 received BA. Mean T1–T2 accuracy declines were 13.9 and 20.0 percentage points, respectively. The within-participant A-minus-B accuracy difference was -1.5 percentage points (95% CI $[-12.1, 9.1]$). These estimates describe form and order sensitivity; the interval does not establish equivalence of the forms.

Leave-one-participant-out checks. Each refit recalculated HC3 standard errors, residual degrees of freedom, and the two-contrast Bonferroni correction. All adjusted differences were positive, ranging from 0.084 to 0.113 for RMW–Notes and from 0.068 to 0.099 for RMW–AI Summary. Adjusted p values ranged from .008 to .061 for RMW–Notes (32/33 refits below .05) and from .023 to .093 for RMW–AI Summary (19/33 refits below .05). These checks describe sensitivity to individual observations rather than independent replications.

Component profiles. Across the six component means, explicit trace coverage and T2 retention had Spearman’s $\rho = .314$ and Pearson’s $r = .372$; these are descriptive associations among six aggregate profiles.

Reference-text length sensitivity. We counted non-whitespace characters, including punctuation and numerals, in the 252 exported Chinese reference propositions. These texts are the reference units available for analysis, not verified verbatim copies of unsplit T1 responses. We fitted a weighted linear model of normalized T2 retention with participant fixed effects, categorical component indicators, and log character count. Each proposition received weight $1/m_{ic}$, where m_{ic} is the number of propositions for participant i in component c , so each participant–component unit had equal total

weight. Covariance estimates were clustered by participant, with the finite-sample factor $[G/(G-1)][(N-1)/(N-K)]$, where $G = 33$, $N = 252$, and $K = 39$. Marginal 95% intervals use a t reference with 32 degrees of freedom and are not corrected for multiple comparisons.

Adjusted means set log length to its weighted sample mean and average equally over participants. The descriptive ordering was unchanged (Table 10). The log-length coefficient was -0.168 (95% CI $[-0.555, 0.220]$). This analysis does not establish that length has no association with retention or that component differences are independent of qualification burden. It is a sensitivity check on the exported text, not a validation of response complexity or segmentation.

Table 10. Exploratory adjustment for reference-text length. Mean character count and unadjusted retention average within participant first; adjusted means come from the participant-fixed-effects model. Intervals are marginal 95% participant-clustered intervals.

Component	Characters	T2	Adjusted	95% CI
Goal	48.7	0.818	0.853	[0.737, 0.970]
Current Judgment and Rationale	41.8	0.515	0.524	[0.427, 0.621]
Active Constraint	35.2	0.697	0.678	[0.570, 0.786]
Rejected Path and Rationale	41.5	0.788	0.797	[0.690, 0.903]
Unresolved Uncertainty	35.9	0.553	0.537	[0.459, 0.615]
Next Action	35.5	0.614	0.596	[0.501, 0.690]

G Supplementary Recovery-Material Diagnostics

G.1 Scoring Procedure

Representation consistency was scored for each T1 reference proposition using the selected recovery material: RMW card statements, AI Summary sentences, or participant-authored notes. Scores were 0 for absent or conflicting content, 1 for partial agreement, and 2 for substantial agreement in essential content and epistemic status. Each reference proposition contributed one score, which was halved and averaged within participant before averaging across participants. The RMW measure concerns card content; it does not measure the combined consistency of the resume brief, cards, and relationship network available during support review.

A task-prior-only reconstruction, generated without participant-specific history, was scored against each participant's T1 propositions using the same rubric and aggregation. It assesses how much agreement may arise from shared task information alone. We report this diagnostic separately from displayed-material consistency.

G.2 Consistency with the Endorsed Reference

Based on the final consistency codes, displayed-material agreement with T1 averaged 0.695 for RMW cards, 0.539 for Self-authored Notes, and 0.704 for AI Summary (Table 11). These means describe the selected materials scored in each condition; the RMW estimate excludes its prose brief and network. The task-prior-only reconstruction, generated without participant-specific history, averaged 0.414 (95% CI $[0.370, 0.458]$). Its agreement was highest for Goal (0.811) and much lower for Unresolved Uncertainty (0.227) and Next Action (0.235; Table 1). Agreement with the endorsed state therefore also arose from shared task information, particularly for the goal.

Table 11. Consistency of the selected displayed materials with T1. The RMW measure scores card statements, not the full three-view material. Means and 95% confidence intervals weight participants equally ($n = 11$ per condition); m is the number of reference propositions scored.

Material	Display consistency [95% CI]	m
RMW cards	0.695 [0.598, 0.793]	82
Self-authored Notes	0.539 [0.439, 0.639]	87
AI Summary	0.704 [0.640, 0.768]	83